|INCH-POUND|

MIL-M-28787/329
AMENDMENT 1
30 July 1993

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

FLIP-FLOP, JK, DIGITAL, KEY CODE QHK

This amendment forms a part of Military Specification MIL-M-28787/329(EC), dated 17 August 1983, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGES 4, 5 AND 6

TABLE I: Delete and substitute new table I as printed on pages 3, 4, 5 and 6 of this amendment.

PAGE 7

4.4.2: Delete and substitute the following:

> "4.4.2 Order of testing. Test conditions A through BB shall be performed in any order with the following exceptions. Test conditions A, B, C, D, and F are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, C, D, or F are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, and F where applicable, followed by the remaining tests in the sequence."

4.5.1, line 5: Delete "D through Y" and substitute "E, G through Y".

4.5.3, line 3: Delete "D through Y" and substitute "E, G through Y".

4.5.3, line 6: Delete "D through BB" and substitute "E, G through BB".

PAGES 47, 48 AND 49

TABLE XV: Delete and substitute new table XV as printed on pages 7, 8, 9, 10, 11, 12 and 13 of this amendment.

AMSC N/A FSC 5963
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PAGE 67

FIGURE 30, ITEM C01 box: Delete "C01" and substitute "C06".

FIGURE 30, NOTE 3, first line: Delete "C01" and substitute "C06".

FIGURE 30, NOTE 3, second line: Delete "Frequency = 120 Hz." and substitute "Frequency = 1 KHz."

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - NW

(Project 5963-0252)

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/ 3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation Initial End-of-Life	III III	S S	10 1	INF INF	10 1	INF INF	10 1	INF INF	10 1	INF INF	10 1	INF INF	MOhm MOhm
B	VCC stress	IV	S	<----------Module must pass all remaining tests---------->										
C1,C2, C24-C26	Input breakdown current	V	S	0	2	0	2	0	2	0	2	0	2	mA
C3-C22, C27-C30	Input breakdown current	V	S	0	1	0	1	0	1	0	1	0	1	mA
C23	Input breakdown current	V	S	0	4	0	4	0	4	0	4	0	4	mA
D1-D14	Output short circuit current	VI	S	-40	-100	-40	-100	-40	-100	-40	-100	-40	-100	mA
E1-E14	Output leakage current	VII	X	0	1	0	1	0	1	0	1	0	1	mA
F1-F24	Input clamp diode voltage	VIII	S	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	V
G1-G8	Functional	IX	X	<----------Output must be as specified in table IX---------->										
H1-H14	Output voltage, low level	X	X	0	500	0	500	0	500	0	500	0	500	mV
J1-J14	Output voltage, high level	X	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
K1-K3, K25,K26	Input current, high level	XI	X	0	200	0	200	0	200	0	200	0	200	µA
K4-K8, K27-K30	Input current, high level	XI	X	0	100	0	100	0	100	0	100	0	100	µA
K9-K24,	Input current, high level	XI	X	0	50	0	50	0	50	0	50	0	50	µA

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C±5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
L1-L3	Input current, low level	XII	X	-0.1	-8.0	-0.1	-8.0	-0.1	-8.0	-0.1	-8.0	-0.1	-8.0	mA
L4	Input current, low level	XII	X	-0.1	-4.0	-0.1	-4.0	-0.1	-4.0	-0.1	-4.0	-0.1	-4.0	mA
L5-L8	Input current, low level	XII	X	-0.10	-4.32	-0.10	-4.32	-0.10	-4.32	-0.10	-4.32	-0.10	-4.32	mA
L9-L24	Input current, low level	XII	X	-0.10	-2.16	-0.10	-2.16	-0.10	-2.16	-0.10	-2.16	-0.10	-2.16	mA
M	Power supply current	XIII	X	0.1	262.0	0.1	262.0	0.1	262.0	0.1	262.0	0.1	262.0	mA
N1-N14	Propagation delay time from clock (tPLH), CL = 50 pF	XIV	S	3	13	3	13	3	15	3	13	3	17	ns
P1-P14	Propagation delay time from clock (tPHL), CL = 50 pF	XIV	S	4	15	4	15	4	17	4	15	4	19	ns
Q1-Q8	Propagation delay time from set (tPLH), CL = 50 pF	XV	S	2	14	2	14	2	16	2	14	2	18	ns
Q9-Q16	Propagation delay time from set (tPLH), CL = 50 pF	XV	S	2	14	2	14	2	16	2	14	2	18	ns
Q17-Q24	Propagation delay time from set (tPLH), CL = 150 pF	XV	Q	2	16	2	16	2	18	2	16	2	20	ns
R1-R6	Propagation delay time from set (tPHL), CL = 50 pF	XV	S	2	21	2	21	2	23	2	21	2	25	ns
R7-R12	Propagation delay time from set (tPHL), CL = 50 pF	XV	S	2	21	2	21	2	23	2	21	2	25	ns
R13-R18	Propagation delay time from set (tPHL), CL = 150 pF	XV	Q	2	23	2	23	2	25	2	23	2	27	ns

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See footnotes at end of table.

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TABLE I Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
S1-S6	Propagation delay time from clear (tPLH), CL = 50 pF	XV	S	2	14	2	14	2	16	2	14	2	18	ns
S7-S12	Propagation delay time from clear (tPLH), CL = 50 pF	XV	S	2	14	2	14	2	16	2	14	2	18	ns
S13-S18	Propagation delay time from clear (tPLH), CL = 150 pF	XV	Q	2	16	2	16	2	18	2	16	2	20	ns
T1-T8	Propagation delay time from clear (tPHL), CL = 50 pF	XV	S	2	21	2	21	2	23	2	21	2	26	ns
T9-T16	Propagation delay time from clear (tPHL), CL = 50 pF	XV	S	2	21	2	21	2	23	2	21	2	26	ns
T17-T24	Propagation delay time from clear (tPHL), CL = 150 pF	XV	Q	2	23	2	23	2	25	2	23	2	27	ns
U1-U14	Operating frequency and minimum clock pulse width check, CL = 50 pF	XVI	S	90	110	90	110	90	110	90	110	90	110	ns
U15-U28	Operating frequency and minimum clock pulse width check, CL = 50 pF	XVI	S	20	30	20	30	20	30	20	30	20	30	ns
V1-V14	Output crosstalk, low level	XVII	Q	0	400	0	400	0	400	0	400	0	400	mV
W1-W14	Output crosstalk, high level	XVII	Q	0	400	0	400	0	400	0	400	0	400	mV
X	Power supply transient amp.	XVIII	Q	0	400	0	400	0	400	0	400	0	400	mV
Y	Power supply filter cap.	XIX	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF

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See footnotes at end of table.

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Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C±5°C		55°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Z1	Input capacitance	XX	Q	5	60	5	60	5	60	6	60	5	60	pF
Z2-Z4	Input capacitance	XX	Q	5	30	5	30	5	30	6	30	5	30	pF
Z5-Z24	Input capacitance	XX	Q	5	15	5	15	5	15	6	15	5	15	pF
AA1-AA28	Setup time	XXI	S	80	120	80	120	80	120	80	120	80	120	ns
BB1-BB28	Hold time	XXII	S	80	120	80	120	80	120	80	120	80	120	ns

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only
4/ Unless otherwise specified, limits are for initial and end-of-life

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TABLE XV. Propagation delay time (tPLH) and tPHL) from set and clear and minimum set and clear pulse width check test.

Test condition	Parameter	Connect			Initial limits 25°C ± 5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q1	Q1 Propagation delay time from set (tPLH), CL = 50 pF	30	19	5	2	14	ns	20,21
Q2	Q2			23				
Q3	Q3			29				
Q4	Q4			26				
Q5	Q5			13				
Q6	Q6			11				
Q7	Q7			18				
Q8	Q8			35				
Q9	Q1			5				
Q10	Q2			23				
Q11	Q3			29				
Q12	Q4			26				
Q13	Q5			13				
Q14	Q6			11				
Q15	Q7			18				
Q16	Q8 Propagation delay time from set (tPLH), CL = 50 pF	30	19	35	2	14	ns	20,21

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TABLE XV. Propagation delay time (tPLH) and tPHL) from set and clear and minimum set and clear pulse width check test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ± 5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
Q17	Q1	Propagation delay time from set (tPLH), CL = 150 pF	30	19	5	2	16	ns	20,21
Q18	Q2				23				
Q19	Q3				29				
Q20	Q4				26				
Q21	Q5				13				
Q22	Q6				11				
Q23	Q7				18				
Q24	Q8	Propagation delay time from set (tPLH), CL = 150 pF			35		16		
R1	$\overline{Q2}$	Propagation delay time from set (tPHL), CL = 50 pF			21		21		
R2	$\overline{Q3}$				8				
R3	$\overline{Q5}$				33				
R4	$\overline{Q6}$	Propagation delay time from set (tPHL), CL = 50 pF	30	19	32	2	21	ns	20,21

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Test condition	Parameter	Connect			Initial limits 25°C ± 5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R5	$\overline{Q7}$ Propagation delay time from set (tPHL), CL = 50 pF	30	19	38	2	21	ns	20,21
R6	$\overline{Q8}$			36				
R7	$\overline{Q2}$			21				
R8	$\overline{Q3}$			8				
R9	$\overline{Q5}$			33				
R10	$\overline{Q6}$			32				
R11	$\overline{Q7}$			38				
R12	$\overline{Q8}$ Propagation delay time from set (tPHL), CL = 50 pF			36		21		
R13	$\overline{Q2}$ Propagation delay time from set (tPHL), CL = 150 pF			21		23		
R14	$\overline{Q3}$			8				
R15	$\overline{Q5}$			33				
R16	$\overline{Q6}$ Propagation delay time from set (tPHL), CL = 150 pF	30	19	32	2	23	ns	20,21

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TABLE XV. Propagation delay time (tPLH) and tPHL) from set and clear and minimum set and clear pulse width check test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ± 5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
R17	$\overline{Q7}$	Propagation delay time from set (tPHL), CL = 150 pF	30	19	38	2	23	ns	20,21
R18	$\overline{Q8}$	Propagation delay time from set (tPHL), CL = 150 pF	30	19	36		23		20,21
S1	$\overline{Q2}$	Propagation delay time from clear (tPLH), CL = 50 pF	19	30	21		14		20,22
S2	$\overline{Q3}$				8				
S3	$\overline{Q5}$				33				
S4	$\overline{Q6}$				32				
S5	$\overline{Q7}$				38				
S6	$\overline{Q8}$				36				
S7	$\overline{Q2}$				21				
S8	$\overline{Q3}$				8				
S9	$\overline{Q5}$				33				
S10	$\overline{Q6}$				32				
S11	$\overline{Q7}$				38				
S12	$\overline{Q8}$	Propagation delay time from clear (tPLH), CL = 50 pF	19	30	36	2	14	ns	20,22

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TABLE XV. Propagation delay time (tPLH) and (tPHL) from set and clear and minimum set and clear pulse width check test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ± 5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
S13	Q2 Propagation delay time from clear (tPLH), CL = 150 pF	19	30	21	2	16	ns	20,22
S14	Q3			8				
S15	Q5			33				
S16	Q6			32				
S17	Q7			38				
S18	Q8 Propagation delay time from clear (tPLH), CL = 150 pF			36		16		
T1	Q1 Propagation delay time from clear (tPHL), CL = 50 pF			5		21		
T2	Q2			23				
T3	Q3			29				
T4	Q4			26				
T5	Q5			13				
T6	Q6			11				
T7	Q7			18				
T8	Q8 Propagation delay time from clear (tPHL), CL = 50 pF	19	30	35	2	21	ns	20,22

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TABLE XV. Propagation delay time (tPLH) and tPHL) from set and clear and minimum set and clear pulse width check test - Continued.

Test condition		Parameter	Connect			Initial limits 25°C ± 5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
T9	Q1	Propagation delay time from clear (tPHL), CL = 50 pF	19	30	5	2	21	ns	20,22
T10	Q2				23				
T11	Q3				29				
T12	Q4				26				
T13	Q5				13				
T14	Q6				11				
T15	Q7				18				
T16	Q8	Propagation delay time from clear (tPHL), CL = 50 pF			35		21		
T17	Q1	Propagation delay time from clear (tPHL), CL = 150 pF			5		23		
T18	Q2				23				
T19	Q3				29				
T20	Q4				26				
T21	Q5	Propagation delay time from clear (tPHL), CL = 150 pF	19	30	13	2	23	ns	20,22

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TABLE XV. Propagation delay time (tPLH) and tPHL) from set and clear and minimum set and clear pulse width check test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ± 5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
T22	Q6	Propagation delay time from clear (tPHL), CL = 150 pF	19	30	11	2	23	ns	20,22
T23	Q7				18				
T24	Q8	Propagation delay time from clear (tPHL), CL = 150 pF	19	30	35	2	23	ns	20,22

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